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Learning Fitness Dynamics of SARS-CoV-2

Stepwise Multinomial Logistic Regression to estimate fitness parameters of viral variants

Lab Rotation Report

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Abstract

Understanding the fitness dynamics of SARS-CoV-2 variants is critical for tracking viral evolution and informing public health decisions. In this work, we present a stepwise maximum likelihood framework based on Multinomial Logistic Regression (MLR) to estimate the fitness of viral variants from genomic surveillance data. Our approach employs the Broyden-Fletcher-Goldfarb-Shanno (BFGS) optimization algorithm in a sliding window fashion, enabling the scalable inference of variant fitness across thousands of lineages over extended time periods. Through simulation experiments, we validate our method against existing tools such as `evofr`, demonstrating superior accuracy and computational efficiency, with a 300-fold reduction in runtime. Applying our approach to SARS-CoV-2 sequencing data from the United States, we recover fitness trajectories for both Nextstrain clades and Pango lineages. These trajectories reveal consistent increases in fitness over time and capture major epidemiological shifts, including the emergence of Omicron. We further link fitness changes to amino acid substitutions, identifying key mutations such as *S:F486P* with strong impacts on viral fitness and tracing their phase-specific relevance throughout the pandemic. Our results show that stepwise optimization is a powerful and efficient method for large-scale fitness estimation and downstream molecular analyses. It offers a computationally efficient tool for real-time genomic surveillance and evolutionary insights into viral adaptation.

Contents

Introduction

SARS-CoV-2 is the most extensively sequenced virus in history, enabling researchers to observe a pandemic at unprecedented resolution [1]. Numerous efforts have been made to classify the resulting genetic diversity, including high-level clustering into Nextstrain clades [2, 3] and more granular classification into Pango lineages [4]. While the Nextstrain system currently recognizes 48 clades, the Pango system comprises thousands of distinct lineages. This complexity underscores the need for scalable methods capable of efficiently analyzing large-scale genomic data.
When new viral variants emerge, they often gain a fitness advantage, typically characterized by increased transmissibility or the ability to evade immune responses. For SARS-CoV-2, such variants have been designated as Variants of Concern (VOCs), as their emergence can drive new waves of infection [5, 6]. From a public health perspective, it is therefore crucial to monitor and track these emerging variants. Estimating viral fitness plays a key role in forecasting potential surges in case numbers and potentially adapting vaccination strategies [7].

To support such efforts, various modeling approaches have been developed to infer variant fitness from genomic surveillance data. Among these, Multinomial Logistic Regression (MLR) has become a widely used approach for modeling the emergence and spread of SARS-CoV-2 variants [8]. Several tools have been developed to estimate variant fitness based on MLR models, some of which are actively used in epidemiological surveillance to assess the risk posed by specific lineages [9]. Many of these models rely on Bayesian inference, employing either variational inference or Markov Chain Monte Carlo (MCMC) methods [10, 11, 12]. However, such approaches often result in high computational costs, limiting the number of variants and the temporal resolution that can be included in the analysis. Alternative methods have been proposed, that use numerical optimization techniques, similar to stochastic gradient descent, to directly maximize the likelihood function [13, 14, 15, 16]. However, even these approaches face scalability challenges, particularly when attempting to estimate the fitness of many variants simultaneously.
Addressing these scalability issues is important for comprehensive and temporally consistent fitness estimation. Estimating the fitness of all variants throughout the course of a pandemic enables consistent comparisons over time and supports downstream analyses. For example, it allows researchers to assess the impact of specific amino acid substitutions at different stages of the pandemic. This highlights the continued need for methods capable of efficiently estimating the fitness of thousands

of variants simultaneously.

Here, we introduce a method for efficiently estimating the fitness advantage of thousands of viral variants using a stepwise Maximum Likelihood approach. Our method is designed for low runtime while maintaining accuracy. We validate its performance through simulation experiments and compare the results to those obtained with the existing tool `evofr` [17]. As a next step, we apply our approach to SARS-CoV-2 sequencing count data spanning the entire course of the pandemic in the United States, enabling downstream analyses of amino acid substitutions impacting fitness.

Methods

Multinomial Logistic Regression Model of Relative Variant Frequencies

We model competing variants with a Multinomial Logistic Regression framework based on "survival-of-the-fittest" selection dynamics [18]. Each of the n distinct variants is characterized by a fitness f_i and an inital frequency $p_i(0)$. The dynamics of the relative frequency of variant i are governed by its fitness advantage compared to the mean population fitness ϕ:

$$\frac{d}{dt}p_i(t) = p_i(t)\left(f_i - \sum_{j=1}^{n} p_j(t)f_j\right) = p_i(t)(f_i - \phi) \tag{1}$$

Solving this differential equation and expressing the initial frequency in logarithmic form leads to the following multinomial logistic regression function, where o_i denotes the log-initial frequency:

$$p_i^{(t)} = \frac{p_i^{(0)}e^{f_i t}}{\sum_{j=1}^{n} p_j^{(0)}e^{f_j t}} = \frac{e^{f_i t + o_i}}{\sum_{j=1}^{n} e^{f_j t + o_j}} \tag{2}$$

Since the model is invariant under scaling of all fitness values or all log-inital frequencies, we must fix the parameters of one variant to ensure identifiability. Without loss of generality, we select the first variant as a pivot and fit its fitness f_1 and log-inital frequency o_1 to zero. This establishes a reference point against which the relative fitness and frequencies of all other variants are estimated.

In this formulation, changes in variant frequency are driven purely by selection advantages. Mutation between variants is not modeled explicitly. Consequently, newly

emerging variants are assumed to already exist at the beginning of the modeled time frame, although with very low initial frequencies.
In clinical sequencing data, we do not directly observe variant frequencies but instead observe variant counts per unit time (e.g., per day), which are subject to stochastic effects. We include this into the model, by assuming sequencing counts as drawn from a multinomial distribution parameterized by the variant frequencies derived above (2). This leads to the following likelihood function for the combined model:

$$\mathcal{L}^{(t)}(s,o) = \mathbb{P}(c^{(t)} \mid p^{(t)}) = \frac{N_t!}{\prod_{i=1}^{n} c_i^{(t)}!} \prod_{i=1}^{n} p_i^{(t)^{c_i^{(t)}}} = \frac{N_t!}{\prod_{i=1}^{n} c_i^{(t)}!} \prod_{i=1}^{n} \left(\frac{e^{f_i t + o_i}}{\sum_{j=1}^{n} e^{f_j t + o_j}} \right)^{c_i^{(t)}} \tag{3}$$

To estimate model parameters, we consider all observed time points and take the logarithm of the likelihood to facilitate optimization. This yields the following log-likelihood function:

$$\begin{aligned} \log \mathcal{L}(s,o) &= \sum_{t=0}^{T} \left(\sum_{i=1}^{n} c_i^{(t)} \cdot \log p_i(t) \right) + \text{const.} \\ &= \sum_{t=0}^{T} \sum_{i=1}^{n} c_i^{(t)} \cdot \left[f_i t + o_i - \log \left(\sum_{j=1}^{n} e^{f_j t + o_j} \right) \right] + \text{const.} \end{aligned} \tag{4}$$

Maximization using Broyden-Fletcher-Goldfarb-Shanno algorithm

We maximize the log-likelihood using the Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm, a quasi-Newton method for numerical optimization. Unlike simple gradient descent, BFGS incorporates curvature information by approximating the Hessian of the objective function, allowing for faster and more stable convergence. The gradient of the log-likelihood with respect to the selection coefficients f_i and log-initial frequencies o_i is given by:

$$\frac{\partial \log \mathcal{L}(f,o)}{\partial f_i} = \sum_{t=0}^{T} c_i^{(t)} \cdot t - \sum_{t=0}^{T} N(t) \cdot \frac{e^{f_i t + o_i} \cdot t}{\sum_{j=1}^{n} e^{f_j t + o_j}} \tag{5}$$

$$\frac{\partial \log \mathcal{L}(f,o)}{\partial o_i} = \sum_{t=0}^{T} c_i^{(t)} - \sum_{t=0}^{T} N(t) \cdot \frac{e^{f_i t + o_i}}{\sum_{j=1}^{n} e^{f_j t + o_j}} \tag{6}$$

This procedure yields maximum likelihood estimates of the variant finesses f_i and their corresponding log-initial frequencies o_i. Additionally, we compute the Hessian matrix of the log-likelihood analytically to quantify uncertainty in parameter estimates and derive confidence intervals using the curvature of the likelihood surface.

Stepwise Optimization Approach to Upscale the Number of Variants

As described in the MLR model above, variants emerge at specific time points and are subsequently replaced by fitter variants. The model assumes that once a variant disappears, it does not reappear at a later time point, a simplification that aligns well with empirical observations. This property can be leveraged to facilitate fitness estimation.

Rather than estimating the fitness parameters for all variants simultaneously, we adopt a stepwise optimization strategy based on a sliding window approach. In each step, parameters are estimated for a subset of variants within a fixed-sized window. Both the window size and the overlap between successive windows can be adapted to the characteristics of the data set. Due to this overlap, some variants are optimized multiple times. For each variant, we retain the parameter estimates from the last window in which it appears as the final estimate. Estimates from earlier windows serve as initial values for the subsequent optimization step.

Each optimization is performed using the complete likelihood function, where parameters of previously estimated variants are fixed to their current values, and parameters of future variants are initialized to zero.

This stepwise procedure enables the simulatenous estimation of fitness for a larger number of variants throughout a broader time range.

Simulation Experiments

To evaluate the performance of the two optimization algorithms, we simulated viral variant count data. The simulation begins with a single variant present at time $t = 0$, whose fitness and log-initial frequency parameters are set to zero. This variant serves as the pivot.

At each subsequent time point, the number of newly emerging variants is drawn from a Poisson distribution with a specified rate. For each new variant, a fitness advantage is sampled from a uniform distribution between 0 and a predefined maximum growth advantage. Its fitness value is determined by adding this growth advantage to the current mean population fitness ϕ. In this study, the maximum growth advantage was set to 0.2. The log-initial frequency of each new variant is then computed such that its frequency at the time of emergence is 10^{-4}.
This procedure allows for the simulation of a variable number of variants across time. The resulting fitness parameters and initial frequencies are used to compute variant frequencies at each time point according to the MLR model. These frequencies are subsequently converted into observed counts by sampling from a multinomial distribution with a specified sampling size per time point.
Datasets are simulated under varying conditions, including different numbers of variants, sampling sizes, and variant emergence rates.

To evaluate algorithm performance on these simulated datasets, it is important to account for the dependence of fitness estimates on earlier variants. Instead of directly comparing estimated and true fitness values, we compute differences between successive parameters for both the true and estimated values and compare these differences. As a final performance metric, we report the mean squared error (MSE) across all such differences.

SARS-CoV-2 data set

To further assess the performance of the stepwise and global BFGS algorithms, we applied them to a SARS-CoV-2 dataset obtained from CoV-Spectrum [19]. This dataset provides daily count data for SARS-CoV-2 variants in the United States, spanning from January 2020 to April 2025. The counts are categorized by Pango lineage and Nextstrain clades.

Esimate the Impact of Amino Acid Substitutions on Fitness over Time

In the next step of the analysis, we use the obtained fitness estimates to predict the impact of specific amino acid substitutions on fitness changes over time. To achieve this, we consider the difference in fitness between the child and parent lineages, denoted as Δs, as defined by the Pango naming scheme. Δs is a vector containing entries for each of the n lineages, where the parent lineage is also present in the dataset. This difference in fitness can be modeled as the sum of the impacts of the amino acid substitutions that led to the formation of the child lineage:

$$\Delta s \approx W \cdot a \tag{7}$$

Here, W is the binary assignment matrix, which assigns all substitutions between a child and parent lineage to each child lineage. This matrix has dimensions $n \times m$. The vector a describes the impact on fitness of all m possible amino acid substitutions. To determine the best-fitting substitution impact vector a, we define a loss function $\mathcal{L}$, that minimizes the deviation between Δs and Wa:

$$\mathcal{L}(a) = (\Delta s - Wa)^{\top} C^{-1} (\Delta s - Wa) + \lambda \|a\|_1 \tag{8}$$

This loss function includes an $l1$-regularization term with regularization parameter λ, based on the assumption that many amino acid substitutions have no impact on fitness. C is the covariance matrix of Δs. The optimization problem is solved using the OSQP solver [20], implemented in the `cvxpy` package.
To quantify the uncertainty of the estimated amino acid substitution effects, we employ a bootstrapping approach. In each bootstrap iteration, Pango Lineages are sampled with replacement, drawing the same number of lineages as in the original dataset. We perform 30 bootstrap runs and average the results to obtain robust estimates of the fitness impact of each amino acid substitution.
Once we have investigated the general impact of different amino acid substitutions, we want to assess how this impact evolves over time. To do so, we define four distinct phases of the pandemic:

- Alpha to Delta variants (approximately January 2020 to December 2021)
- Omicron (approximately January 2022 to December 2022)
- Omicron XBB (approximately January 2023 to November 2023)
- JN1 (approximately December 2023 to April 2025)

We compare the impact of amino acid substitutions across these phases of the pandemic.

Results

Validation of BFGS algorithm and comparison to evofr on simulated data.

As a proof of concept, both the global and stepwise BFGS algorithms were run on simulated data. To validate their performance against existing methods, they were compared to `evofr` [17], a tool that estimates variant fitness using variational inference rather than numerical likelihood estimation.

The analysis was first performed on simulated datasets containing 100 variants (Figure 1A). On these datasets, both the global and stepwise BFGS algorithms successfully recovered the true fitness values from the count data, performing equally well (Figure 1B and D). In contrast, `evofr` struggled to converge, resulting in a significant underestimation of fitness values (Figure 1C and E).

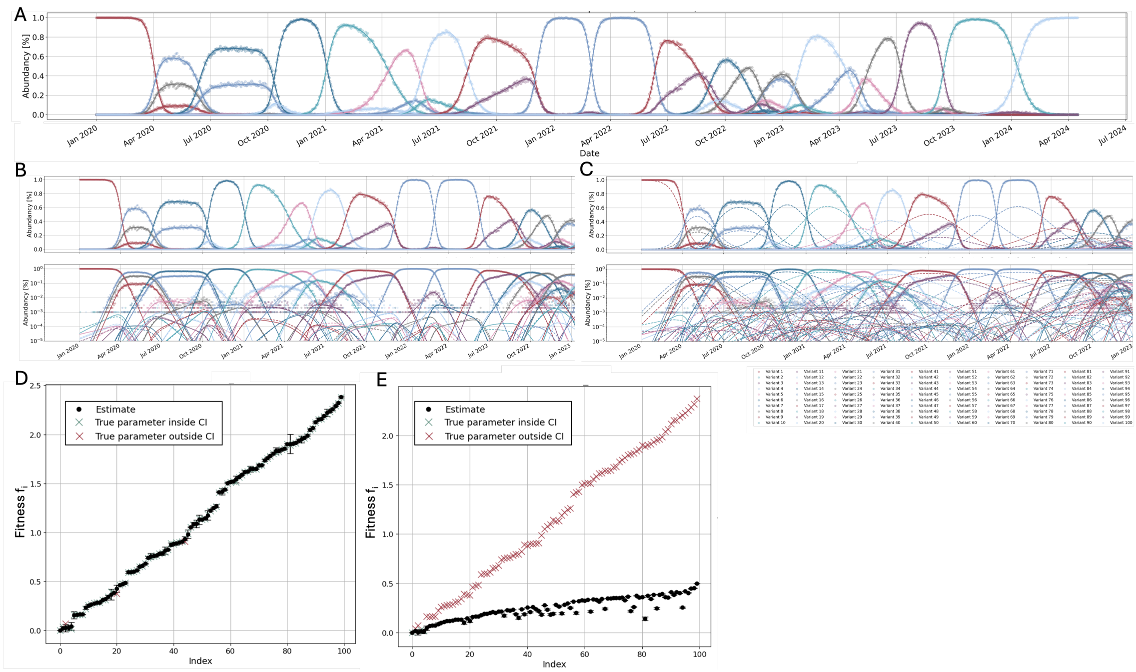

Figure 1: Estimating fitness of simulated variants using the BFGS algorithm and `evofr`. (A) Simulated count and frequency data for 100 viral variants over a period of approximately 4 years. (B) Reconstruction of variant frequencies using fitness values estimated by the stepwise BFGS algorithm based on count data. True frequencies are shown as solid lines, estimated frequencies as dashed lines, on both a linear scale (top) and a log scale (bottom). The global BFGS algorithm yields similar results. (C) Same reconstruction using `evofr` which shows convergence issues and underestimates variant fitness. (D) Fitness estimates with 95% confidence intervals for all variants based on stepwise BFGS. True fitness values are shown as red or green crosses, depending on whether they lie within the confidence intervals. (E) Fitness estimates produced by `evofr`, showing systematic underestimation.

Already for datasets with 100 variants, the runtime difference is substantial: While `evofr` takes approximately 10 minutes, the BFGS algorithms completes in about 2 second, leading to a 300-fold speedup.

Comparison of Global and Stepwise BFGS Algorithm on Simulated Data

We next increased both the number of variants and the overall time span of the analysis by decreasing the variant appearance rate.
On smaller datasets, as shown in Figure 1, the global and stepwise BFGS algorithms perform equally well. However, for larger datasets, the global BFGS algorithm begins to fail, whereas the stepwise approach continues to recover the true fitness parameters (see Figure 2A-C for an example dataset with 500 variants).

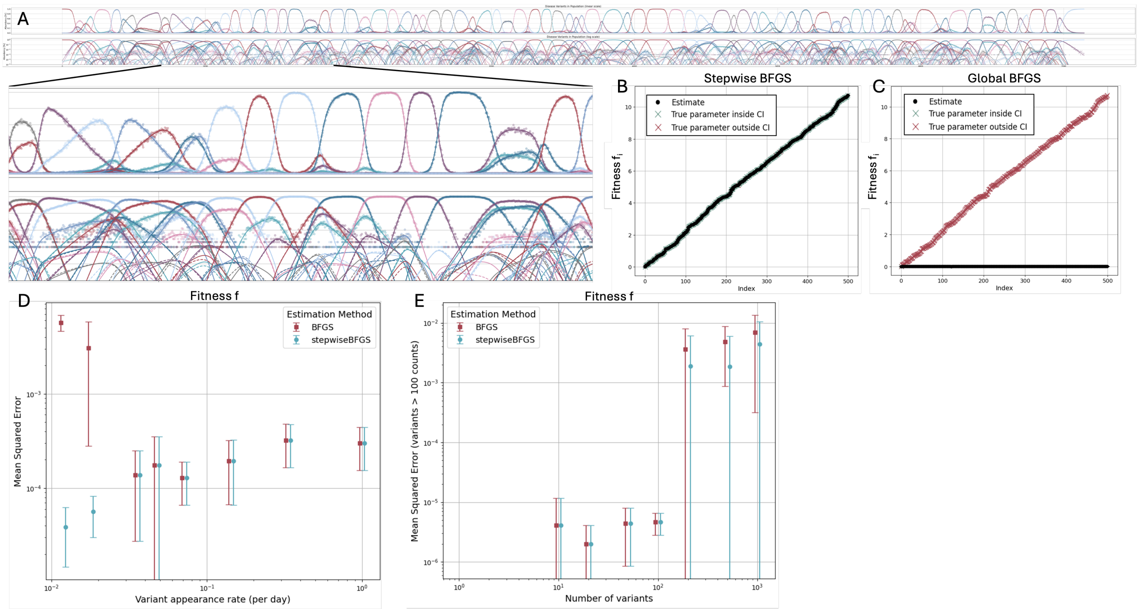

Figure 2: Comparison of global and stepwise BFGS algorithm on multi-variant datasets. (A) Reconstruction of variant frequencies based on variant fitness values estimated with the stepwise BFGS algorithm based on count data. A simulated data set containing 500 variants was used. True frequencies are plotted with a solid line and frequency estimates with a dashed line on a linear scale (top) and on a log scale (bottom). (B) Fitness estimates with confidence intervals for all viral variants, estimated with the stepwise BFGS algorithm. True values are plotted as red or green crosses, depending on whether they lay within the 95% confidence intervals. (C) Fitness estimates obtained with the global BFGS algorithm. The algorithm fails on a high variant dataset. (D) Mean squared error of fitness estimates of all variants in 5 simulated datasets with 50 variants for stepwise and global BFGS algorithm. A decreasing rate of variant appearance leads to a longer overall time range. As a consequence, the global BFGS algorithm starts to fail, whilst the stepwise algorithm can still reproduce the fitness estimates correctly. (E) Mean squared error of fitness estimates of all variants in 5 simulated datasets with at increasing number of variants. At more than 100 variants stepwise and global algorithm fail for some datasets. However, the global algorithm is failing more frequently leading to higher MSEs.

To investigate this further, we applied both algorithms to datasets with varying numbers of variants and variant appearance rates, simulating five datasets per setting. Figures 2D and E show the mean squared error (MSE) across all fitness estimates within these replicates.

When varying the variant appearance rate, we observe that the stepwise BFGS algorithm outperforms the global algorithm for lower appearance rates. This is expected, since lower rates lead to longer overall time spans, which hinder global fitness estimation but affect the stepwise approach far less. Excluding datasets where the global algorithm fails entirely, the MSE increases as the appearance rate rises for both algorithms, indicating that a higher degree of overlapping variants complicates fitness estimation for both methods.

Varying the number of variants instead, we find that starting from about 200 variants, both algorithms exhibit high errors. Closer inspection reveals that while the global algorithm fails completely on these datasets (Figure 2C), the stepwise algorithm still recovers the correct overall fitness trends, but the true values fluctuate around its estimates. Because our error metric is based on differences between successive parameters, these fluctuations still translate into high errors. To address this, in Figure 2E we restrict the error calculation to variants with total counts above 100, which reduces this issue somewhat but does not resolve it entirely.

We further find that fitness estimation error strongly depends on sampling size (see Supplementary Figure 6). Increasing the sampling size reduces estimation errors, as does restricting the analysis to variants with higher counts. Both observations underscore the difficulty of accurately estimating the fitness of low-frequency variants, which provide limited data to inform their growth dynamics.

Overall, despite the limitations of the current error metric, a clear trend emerges: for datasets with more than 100 variants, the stepwise BFGS algorithm substantially outperforms the global BFGS algorithm, successfully recovering fitness trends even in cases where the global algorithm fails. Further work is needed to develop a more robust error metric, but these results already highlight the advantages of the stepwise approach for larger and more complex datasets.

Application to SARS-CoV-2 data

The next step of the analysis involved applying the BFGS algorithms to SARS-CoV-2 count data, grouped at two different levels of resolution: Nextstrain Clades and the more fine-grained Pango Lineages. At the clade level, both the global and stepwise BFGS algorithms yielded similar results (Figure 3A). In both cases, frequency estimates based on the inferred parameters closely matched the observed counts, although for some clades, counts declined slightly faster than the estimated frequencies.

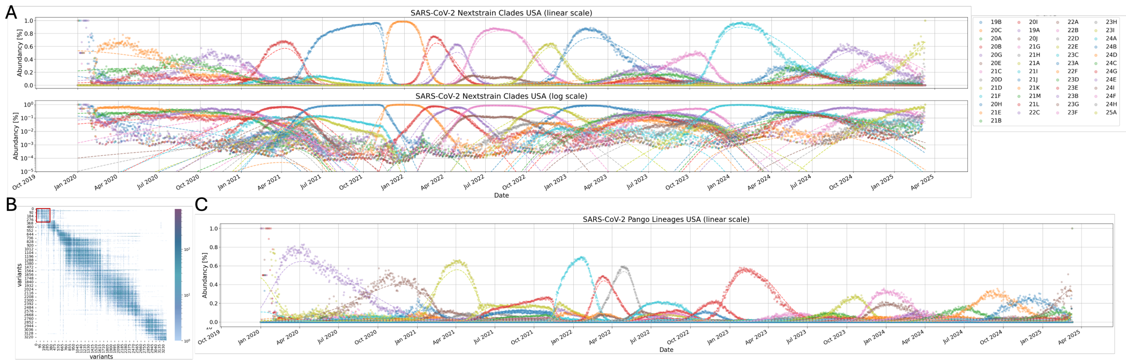

Figure 3: Application of Stepwise BFGS Algorithm to SARS-CoV-2 Nextstrain Clades and Pango Lineages. (A) Frequency estimates (dashed lines) based on estimated variant parameters and observed count data for SARS-CoV-2 Nextstrain Clades from January 2020 to April 2024. Both the Stepwise BFGS (window size 10, overlap 5) and the standard BFGS algorithm produce similar results. Frequencies are shown on both a linear scale (top) and a logarithmic scale (bottom). (B) Co-occurrence matrix of SARS-CoV-2 Pango Lineages, indicating a window size of 300 for the Stepwise BFGS algorithm. (C) Frequency estimates (dashed lines) derived from estimated fitness parameters and count data for SARS-CoV-2 Pango Lineages using the Stepwise BFGS algorithm (window size 300, overlap 50).

At the lineage level, we compared different window sizes for the Stepwise BFGS algorithm and found that a window size of 300 with an overlap of 50 provided the best fit to the count data (Figure 3C). This is also supported by the co-occurrence matrix (Figure 3B). Due to the large number of Pango Lineages, we chose to focus on the stepwise algorithm for this resolution. Comparing the fits between Pango Lineages and Nextstrain Clades, we observed that lineage-level modeling achieved a closer fit to the count data. The residual errors seen at the clade level likely result from aggregating multiple variants into clades, effectively averaging their fitness values and obscuring fine-scale variation.

Fitness Development over time

We further examined the fitness development of Nextstrain Clades and Pango Lineages over time. Plotting fitness against the mean time of occurrence for each variant revealed a general increase in fitness, with a marked jump corresponding to the emergence of Omicron variants. In Figure 4A, Pango Lineages are colored according to their respective Nextstrain Clades, illustrating the close agreement between the two estimation levels and thereby supporting the validity of the stepwise BFGS algorithm.

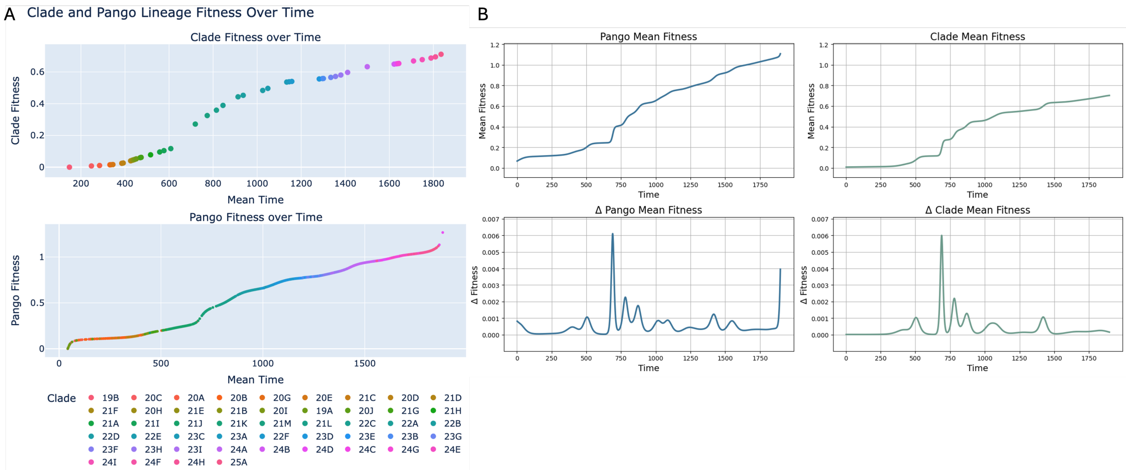

Figure 4: Comparison of Fitness Estimates between Nextstrain Clades and Pango Lineages over Time (A) Fitness estimates of all Nextstrain Clades (top) and Pango Lineages (bottom) plotted at their mean time of occurrence. Variant fitness generally increases over time. Pango Lineages are colored according to their corresponding Nextstrain Clades, highlighting the similar trends in fitness between these two variant groupings. (B) Mean fitness over time for Pango Lineages (blue) and Nextstrain Clades (green), along with their temporal derivatives. Both groupings display similar overall dynamics. However, fitness increases more rapidly for Pango Lineages, suggesting that aggregating lineages into clades masks fine-scale adaptation of lineages.

We also analyzed mean fitness over time for both clades and lineages, along with their temporal derivatives (Figure 4B). Both show similar overall dynamics, characterized by an increase in fitness with some jumps when new waves of infection emerged. However, for Pango Lineages fitness increases more rapidly. This suggests that aggregating lineages into broader clades averages out their individual fitness differences and obscures fine-scale adaptive dynamics.

Impact of Amino Acid Substitutions on Fitness over Time

In the final step of the analysis, we used the Pango Lineage fitness estimates to identify amino acid substitutions with a strong impact on viral fitness. Since many substitutions are expected to have no effect, we selected a regularization parameter λ that maximized the number of substitutions with zero impact, thereby promoting sparsity while keeping the objective function value low. As shown in Figure 5A, $\lambda = 0.1$ optimally balances these criteria.

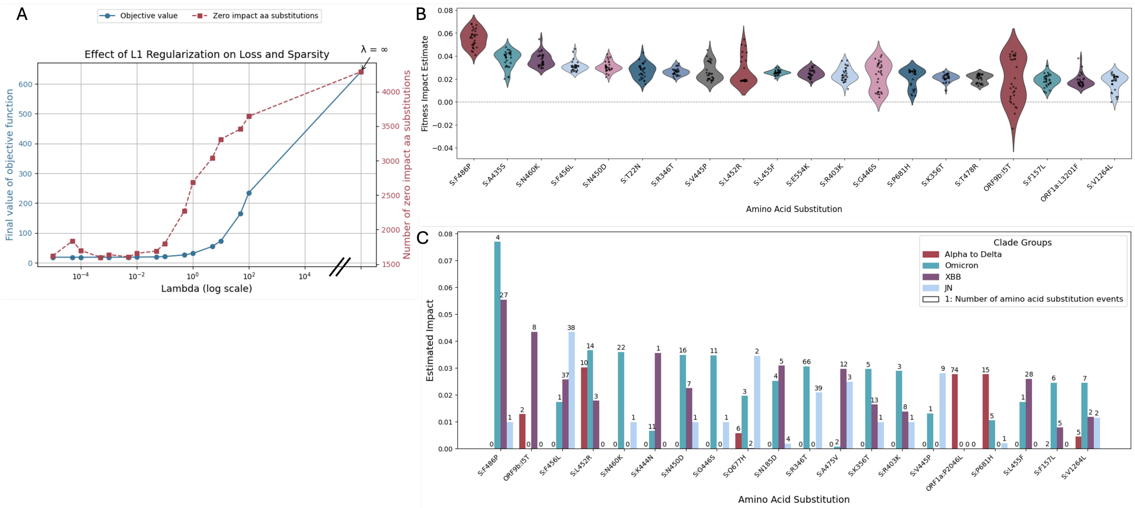

Figure 5: Impact of Amino Acid Substitutions on Variant Fitness over Time. (A) Objective function value (left axis) and sparsity of the solution, measured as the number of amino acid substitutions with zero estimated impact (right axis), plotted against the regularization parameter λ. A value of $\lambda = 0.1$ maximizes sparsity while maintaining a low objective function value. (B) Bootstrap results for the 20 amino acid substitutions with the highest estimated impact. Only substitutions observed in at least 10 variants are included. The spike protein substitution *F486P* shows the strongest impact. (C) Estimated impact of amino acid substitutions across four phases of the SARS-CoV-2 pandemic: Alpha to Delta, Omicron, XBB and JN1. Only substitutions observed in at least 10 variants are considered. The 20 most impactful substitutions in any phase are displayed, with bar labels indicating the number of variants in which each substitution occurs per phase.

Figure 5B presents the results of the optimization. The 20 amino acid substitutions with the highest estimated impact on fitness are displayed. Substitutions occurring in fewer than 10 Pango Lineages were excluded, as their effects could not be reliably inferred. Uncertainty was assessed using a bootstrapping approach. The spike protein substitution *F486P* consistently showed the strongest impact, followed by *A435S* and *N460K*, both also located in the spike protein.
Finally, we examined how the impact of amino acid substitutions evolved over different phases of the pandemic. As shown in Figure 5C, some substitutions, such as *S:R346T*, remained influential across multiple phases. Others showed phase-specific relevance: for example, *ORF1a:P2046L* was primarily associated with the early phases of the pandemic, *S:F486P* was more impactful during the Omicron and XBB phases, and *S:F456L* emerged in later phases.

Discussion

In this work, we introduced a stepwise BFGS algorithm that successfully estimates variant fitness from genomic surveillance data. Our method outperforms existing

approaches such as `evofr` and can efficiently handle datasets comprising thousands of variants spanning extended time periods.
Beyond improved performance, the stepwise BFGS algorithm is also computationally efficient. Additionally, its sequential structure allows new count data to be easily incorporated into existing estimates, making it well-suited for integration into real-time genomic surveillance dashboards. Such tools enable continuous monitoring of circulating variants and provide rapid fitness estimates for newly emerging lineages, thereby informing public health decisions.

Although our algorithm demonstrates strong performance, several limitations occur that need further investigation. During our evaluation, we observed that the global BFGS algorithm is prone to numerical instabilities that can hinder convergence. Fortunately, failures can be easily identified by inspecting how estimated frequencies fit the observed count data, ensuring reliable results for real-world datasets such as the SARS-CoV-2 data analyzed in this study. Nevertheless, future work is needed to better understand and address these issues to improve algorithmic robustness.
Another challenge lies in defining an appropriate evaluation metric. Fitness estimates for later variants depend on those of earlier variants, complicating error assessment. While our use of differences between successive fitness values prevents this dependency, it cannot distinguish between an algorithm that fails entirely (assigning all fitness values to zero) and one that captures the correct trend but fluctuates around true values. Further research is required to develop a more robust error metric.
Finally, we observed that fitness estimation accuracy strongly depends on the number of counts per variant, with low-frequency variants exhibiting poorer fits. However, this limitation is acceptable in practice, since variants with higher counts tend to be the most epidemiologically relevant.

Applying the stepwise BFGS algorithm to SARS-CoV-2 data successfully recovered fitness estimates for both Nextstrain Clades and Pango Lineages. Fitness estimates for Pango Lineages aligned well with their corresponding Nextstrain Clades, supporting the validity of our approach. Across the pandemic, we observed a general increase in fitness, consistent with the expectation that new variants must outcompete their predecessors to spread [8]. The algorithm also captured major epidemiological events, such as the emergence of Omicron approximately two years into the pandemic [21]. Interestingly, fitness increased faster at the lineage level than at the clade level, likely due to fine-scale adaptations being masked when lineages are aggregated into clades. Exploring this discrepancy further could provide deeper insights into the underlying evolutionary dynamics.

Having established consistent fitness estimates across the pandemic, we can further leverage them for downstream analyses of amino acid substitution impacts over time. The spike protein mutation *F486P* emerged as having the highest estimated impact, consistent with its known role in increasing infectivity by enhancing binding affinity to the ACE2 receptor [22], particularly in XBB variants [23, 24]. This aligns well with our timed analysis, where *F486P* was most relevant during the XBB phase. Other high-impact substitutions of the spike protein included *A435S* and *K356T*, both previously identified by Murrell's group as functionally important [25].
Examining the temporal dynamics more closely, we find that mutations identified as impactful during distinct phases of the pandemic correspond well with their documented occurrences in the literature. For example, mutations such as *ORF1a:P2046L* and *S:L452R*, detected as relevant during early phases, are characteristic of the Delta variant [26, 27]. Similarly, *S:R346T*, found to be associated with Omicron and JN1 variants, lies within the receptor-binding domain (RBD) of the spike protein and is linked to enhanced antibody escape in Omicron lineages [28]. More recently, *S:F456L* has been linked to XBB and JN1 lineages and shown to increase ACE2 binding affinity [29, 30]. The consistency between our findings and established literature validates our fitness-based approach for quantifying amino acid substitution impacts.

In summary, our stepwise BFGS algorithm not only efficiently estimates variant fitness estimation but also allows downstream analyses that illuminate the molecular drivers of viral evolution. By connecting fitness dynamics with underlying amino acid changes, our approach offers a powerful tool for tracking viral adaptation and anticipating the emergence of new variants. Future work should focus on refining error metrics, extending the framework to other pathogens, and further integrating fitness estimation into real-time genomic surveillance systems.

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Supplement

Influence of Sampling Size on Fitness Estimates

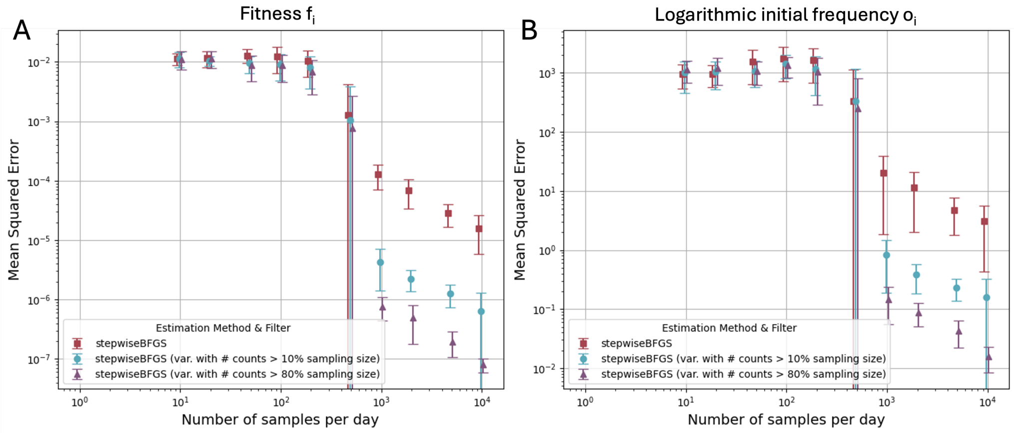

Figure 6: Effect of Sampling Size on accuracy of fit. Average mean squared error of fitness (A) and logarithmic initial frequency (B) estimates of 5 runs per sampling size of the stepwise algorithm. An increase in sampling size significantly reduced the estimation error. Only taking into account variants with more counts than 10% or 80% of the sampling size for error calculation decreases the error further.